

EDUCATION

Kyungpook National University (KNU) Daegu, Republic of Korea
Bachelor of Science in Physics Mar. 2020 – Feb. 2024
Overall GPA: 4.11/4.3 | Major GPA (71 credits): 4.19/4.3

PUBLICATIONS

[4] **Magnetic characterization of electronic components for portable atomic sensors using a zero-field optically pumped magnetometry platform**
Manuscript submitted; arXiv:2609.07270 (2026). DOI: 10.48550/arXiv:2609.07270
H. Kim, S. Lee, S. H. Yim, T. Jeong, and Y. Lim

[3] **Wire-sealed atomic gas cells for noble-gas nuclear magnetic resonance applications**
Manuscript under review
S. H. Hong, Y. Lim, S. H. Yim, J. B. Nam, S. Lee, T. Jeong, and H. Kim

[2] **Magnetic Anomaly Detection of an Undersea Tunnel Using a Drone-Mounted All-Optical Pulsed Atomic Magnetometer**
IEEE Sensors Journal, Early Access (2026). DOI: 10.1109/JSEN.2026.3729190
H. Kim, S. H. Yim, S. Lee, E.-S. Bang, S. R. Yeo, Y. J. Kim, T. Jeong, and Y. Lim

[1] **Aging test of atomic vapor cell with Al₂O₃ wall coating on cubic glass**
Applied Optics **64**, 7932–7937 (2025). DOI: 10.1364/AO.572036
H. Kim, T. Jeong, S. H. Hong, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim

SELECTED PRESENTATIONS

[3] **Probing magnetic noise from portable sensor components using a zero-field optically pumped magnetometer**
Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2026)

[2] **Experimental systems for spectroscopy and zero-field OPM with atomic vapor cells**
Invited oral presentation; KISTI-SNU Joint Workshop (2026)

[1] **Lifetime extension of rubidium vapor cells by Al₂O₃ coating**
Poster presentation; APS Division of Atomic, Molecular, and Optical Physics (DAMOP) Meeting (2025)

RESEARCH EXPERIENCE

Agency for Defense Development (ADD) Daejeon, Republic of Korea
Research Officer for National Defense (Military service) Jun. 2024 – Present
Supervisors: Dr. Sin Hyuk Yim and Dr. Sangkyung Lee
Research in experimental atomic physics and quantum sensing

- Developed an automated absorption spectroscopy system with spectral modeling to characterize atomic vapor cells and contributed to their fabrication
- Built a zero-field optically pumped magnetometry platform to evaluate component-induced magnetic interference and provide design guidance for portable atomic sensors
- Deployed a drone-mounted atomic magnetometer for magnetic anomaly detection and established a quantitative framework for interpreting the acquired data

Kyungpook National University (KNU)	Daegu, Republic of Korea
<i>Undergraduate Researcher</i>	
Novel Applied Nano Optics Lab (Advisor: Prof. Junyeob Yeo) Fabricated a flexible photoelectrochemical cell using laser processing High Energy Physics Lab (Moon Lab) (Advisor: Prof. Chang-Sung Moon) Developed momentum reconstruction algorithms for silicon detector simulations	Jun. 2021 – Jun. 2022 Jun. 2020 – Feb. 2021

HONORS AND AWARDS

KFAS Study Abroad Workshop (National Sciences) Korea Foundation for Advanced Studies (KFAS)	Jul. 2026
Physics Alumni Association Award and Scholarship (Outstanding Graduate) Department of Physics Alumni Association, Kyungpook National University	Feb. 2024
2nd Prize, 2023 MiliTECH Challenge Hosted by KAIST; awarded by the Agency for Defense Development	Dec. 2023
2nd Prize, 2022 MiliTECH Challenge Hosted by KAIST; awarded by the Korea Military Academy	Dec. 2022
3rd Prize, Academic Conference College of Natural Sciences, Kyungpook National University	Dec. 2021
Dean’s List College of Natural Sciences, Kyungpook National University	Nov. 2021, Apr. 2022, Apr. 2023

GRANTS AND SCHOLARSHIPS

National Science and Technology Scholarship Korea Student Aid Foundation (KOSAF)	Mar. 2022 – Feb. 2024
Academic Merit Scholarship (Hyoseok Scholarship) Kyungpook National University Alumni Association	Apr. 2022
Academic Excellence Scholarship Kyungpook National University	Aug. 2020, Feb. 2021, Aug. 2021

OTHER RESEARCH OUTPUTS

Buffer Gas Pressure Estimation for OPMs Registered software; Korea Copyright Commission (C-2024-047156)	Dec. 2024
Rubidium Vapor Cell Lifetime Monitoring and Analysis Tool Registered software; Korea Copyright Commission (C-2024-054414)	Dec. 2024

ADDITIONAL EXPERIENCE

Teaching Assistant, Classical Mechanics II (PHYS212)	Sep. 2023 – Dec. 2023
Teaching Assistant, Electromagnetism II (PHYS312)	Mar. 2023 – Jun. 2023
International Student Tutor, KNU	Feb. 2022 – May 2022

SKILLS

Programming: Python, Wolfram Language	Software: LabVIEW, SolidWorks, COMSOL Multiphysics
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